RACEDAY API ENDPOINT PLAN

Part 1 - API plan for the run / walk / cycle event system. 23 endpoints. Part 2 implements exactly these routes.

Authentication: session-based. POST /api/auth/login verifies the password hash and issues a server-signed HttpOnly session cookie containing the user id and role; every later request is checked against it. Everything except register and login requires a logged-in user (401 otherwise), and role rules return 403 when the wrong role (or a non-owner Organiser) calls an endpoint.

METHOD	ROUTE	DESCRIPTION	ROLE REQUIRED	REQUEST BODY	EXPECTED RESPONSE
AUTHENTICATION					
POST	`/api/auth/register`	Registers a new Organiser or Participant. Participants must give a date of birth; Organisers must give the organiser invite code so the role cannot be self-assigned. Password is stored as a salted hash.	Public	`{ fullName, email, password, role, phone?, dateOfBirth?, organiserCode? }`	201 Created - user summary (no password). 400 invalid data. 403 bad organiser code. 409 email already registered.
POST	`/api/auth/login`	Checks credentials and starts a session: the server sets an HttpOnly cookie holding the user id and role, used for all later requests.	Public	`{ email, password }`	200 OK - user summary + session cookie. 401 invalid credentials.
POST	`/api/auth/logout`	Ends the current session.	Any logged-in user	`None`	204 No Content. 401 not logged in.
USER PROFILE					
GET	`/api/users/me`	Returns the logged-in user's own profile.	Any logged-in user	`None`	200 OK - profile. 401.
PUT	`/api/users/me`	Updates the logged-in user's own profile.	Any logged-in user	`{ fullName, phone?, dateOfBirth? }`	200 OK - updated profile. 400 invalid data. 401.
EVENTS					
GET	`/api/events`	Lists events. Optional filters: eventType (Run/Walk/Cycle), from, to.	Any logged-in user	`Query: eventType?, from?, to?`	200 OK - array of events. 401.
GET	`/api/events/mine`	Lists the events created by the logged-in Organiser.	Organiser	`None`	200 OK - array of events. 401. 403.
GET	`/api/events/{id}`	Returns one event with its categories.	Any logged-in user	`None`	200 OK - event with categories. 401. 404.
POST	`/api/events`	Creates an event (name, description, date, location, distance, type run/walk/cycle).	Organiser	`{ name, description, eventDate, location, distanceKm, eventType }`	201 Created - event. 400. 401. 403.
PUT	`/api/events/{id}`	Updates an event the Organiser created.	Organiser (owner)	`{ name, description, eventDate, location, distanceKm, eventType }`	200 OK. 400. 401. 403 not owner. 404.
DELETE	`/api/events/{id}`	Deletes an event the Organiser created, if nobody is enrolled.	Organiser (owner)	`None`	204 No Content. 401. 403. 404. 409 has enrolments.
CATEGORIES					
GET	`/api/events/{eventId}/categories`	Lists the age/distance categories of an event.	Any logged-in user	`None`	200 OK - array. 401. 404.
POST	`/api/events/{eventId}/categories`	Adds a category. Age: minAge and/or maxAge. Distance: distanceKm.	Organiser (owner)	`{ name, description?, categoryType, minAge?, maxAge?, distanceKm? }`	201 Created. 400 rule broken. 401. 403. 404.
PUT	`/api/categories/{id}`	Updates a category on the Organiser's own event.	Organiser (owner)	`{ name, description?, categoryType, minAge?, maxAge?, distanceKm? }`	200 OK. 400. 401. 403. 404.

METHOD	ROUTE	DESCRIPTION	ROLE REQUIRED	REQUEST BODY	EXPECTED RESPONSE
DELETE	`/api/categories/{id}`	Deletes a category that has no enrolments.	Organiser (owner)	`None`	204 No Content. 401. 403. 404. 409 has enrolments.
EVENT ENROLMENTS					
POST	`/api/events/{eventId}/enrolments`	Enters the logged-in Participant into an event by selecting a category. Records the link between participant, event and category; age categories are checked against the participant's age on event day.	Participant	`{ categoryId }`	201 Created - enrolment. 400 past event / not eligible. 401. 403. 404. 409 already enrolled.
GET	`/api/events/{eventId}/enrolments`	Lists every enrolment for one of the Organiser's events.	Organiser (owner)	`None`	200 OK - array. 401. 403. 404.
GET	`/api/enrolments/mine`	Lists the logged-in Participant's own enrolments.	Participant	`None`	200 OK - array. 401. 403.
DELETE	`/api/enrolments/{id}`	Withdraws the Participant's own enrolment.	Participant (owner)	`None`	204 No Content. 401. 403. 404. 409 result exists.
RESULTS					
POST	`/api/enrolments/{enrolmentId}/result`	Captures finish time and finishing position for an enrolment after the event.	Organiser (owner of event)	`{ finishTime (hh:mm:ss), finishPosition }`	201 Created. 400. 401. 403. 404. 409 result exists / position taken.
PUT	`/api/results/{id}`	Corrects a previously captured result.	Organiser (owner of event)	`{ finishTime, finishPosition }`	200 OK. 400. 401. 403. 404. 409.
GET	`/api/events/{eventId}/results`	Results of an event, ordered by category then finishing position.	Any logged-in user	`None`	200 OK - array. 401. 404.
GET	`/api/results/mine`	The logged-in Participant's own results.	Participant	`None`	200 OK - array. 401. 403.